Analytical Report: Metro Interstate Traffic Volume Analysis

Title: Comprehensive Exploratory Data Analysis of Interstate Traffic Patterns

Role: Lead Data Analyst

Dataset: Metro_Interstate_Traffic_Volume.csv (Hourly interstate 94 Westbound traffic volume for MN DoT)

Task Objective: To extract actionable insights into traffic behavior through statistical summaries, time-series forecasting, and environmental impact analysis.

Libraries Used: Pandas, NumPy, Matplotlib, Seaborn, Scipy.

Prompt: You are a data analyst tasked with performing a complete and in-depth Exploratory Data Analysis (EDA) on the provided Excel dataset containing traffic volume data. The dataset includes features such as date-time information, traffic volume, weather conditions, temperature, holiday indicators, and other relevant attributes. Your goal is to extract meaningful insights, identify patterns, and present findings using a variety of visualizations and statistical summaries.Start by loading and understanding the dataset structure. Display the first few rows and summarize the dataset, including column names, data types, and missing values. Perform data cleaning and preprocessing, which includes handling null values, correcting data types (especially converting date-time columns), removing duplicates, and ensuring consistency in categorical variables.Next, perform univariate analysis on all features. For numerical columns such as traffic volume and temperature, compute descriptive statistics (mean, median, standard deviation, min, max) and visualize distributions using histograms, box plots, and density plots. For categorical variables like weather conditions and holidays, use bar charts and pie charts to understand frequency distributions.Proceed with bivariate and multivariate analysis to uncover relationships between variables. Use.Scatter plots to analyze relationships between temperature and traffic volume.Line charts to study traffic trends over time (hourly, daily, monthly).Heatmaps to visualize correlations between numerical variables.Grouped bar charts to compare traffic volume across different weather conditions or days of the week.Stacked bar charts to analyze combined effects (e.g., weather + time of day).Perform time series analysis by extracting features such as hour, day, month, and year from the date column. Visualize trends using time-series line plots, and identify peak traffic hours, busiest days, and seasonal patterns. Include rolling averages or moving averages to smooth trends.Analyze impact of external factors such as weather and holidays on traffic volume. Use box plots and violin plots to compare distributions under different conditions. Identify anomalies or outliers using box plots and z-score methods, and visualize them clearly.Include advanced visualizations such as:Pair plots for feature relationshipsClustered heatmapsArea charts for cumulative trend.Finally, summarize key insights in a clear and structured manner.

Highlight:Peak traffic periodsInfluence of weather conditionsSeasonal and time-based patternsAny anomalies or unusual observationsEnsure all visualizations are well-labeled with titles, axis labels, and legends. The output should be insightful, visually rich, and suitable for decision-making or dashboard creation.

1. Data Preprocessing & Cleaning

- **Missing Values:** Handled the holiday column where ~99% were null by imputing "None" to represent regular workdays.
- **Duplicate Removal:** Identified and dropped 17 redundant entries to ensure statistical integrity.
- **Feature Engineering:** Parsed date_time to extract hour, day_of_week, month, and year. Created a time_segment categorical variable (Day/Night).
- **Anomaly Handling:** * Filtered out records with temp = 0K (sensor error).
    - Removed extreme rainfall outliers (>9000mm/h) that were physically impossible.

2. Section-Wise Analysis (Breakdown by Visualization)

Graph 1: Univariate Distribution (Histograms & KDE)

- **Description:** Visualized the distribution of Traffic Volume, Temperature, and Cloud Cover.
- **Insight:** Traffic volume is bimodal, indicating two high-intensity periods (rush hours) rather than a normal distribution.

Graph 2: Boxplots for Range Analysis

- **Description:** Statistical range for volume and temperature to identify standard variance and potential outliers.
- **Insight:** Traffic shows high variance but few statistical outliers after cleaning, indicating consistent daily cycles.

Graph 3: Weather Frequency (Count Plot)

- **Description:** Frequency of different weather_main conditions.
- **Insight:** "Clouds" and "Clear" are the dominant weather states, providing a large sample size for reliable correlation.

Graph 4: Temperature vs. Traffic (Scatter Plot)

- **Description:** Investigated the relationship between thermal conditions and volume.
- **Insight:** Weak linear correlation; however, moderate temperatures (280K-290K) show the highest density of peak traffic.

Graph 5: Hourly Traffic Trend (Line Chart)

- **Description:** Mean traffic volume mapped across 24 hours.
- **Insight:** Two distinct peaks identified at 07:00 and 16:00. The evening peak is broader and

heavier than the morning peak.

Graph 6: Monthly Traffic Seasonality (Line Chart)

- **Description:** Average volume per month.
- **Insight:** Traffic increases significantly in the warmer months (June-August) and dips in mid-winter (January), likely due to weather hazards.

Graph 7: Correlation Heatmap

- **Description:** Matrix showing Pearson correlation coefficients between all numerical features.
- **Insight:** Hour shows the strongest correlation with volume, whereas rain_1h shows negligible immediate impact on volume totals.

Graph 8: Busiest Days (Bar Chart)

- **Description:** Comparison of traffic across Monday through Sunday.
- **Insight:** Thursday and Friday are the busiest days. Weekend traffic (Sat/Sun) drops by approximately 30-40% compared to the weekday mean.

Graph 9: Weather + Time Segment (Stacked Bar Chart)

- **Description:** Combined effect of weather type and Day/Night segments.
- **Insight:** "Clouds" and "Clear" weather during the day account for the highest cumulative traffic load.

Graph 10: Rolling Average (Time Series)

- **Description:** 30-day moving average of daily traffic volume.
- **Insight:** Smooths out daily volatility to show a steady growth trend over the multi-year observation period.

Graph 11: Holiday Impact (Boxplot)

- **Description:** Comparing distributions of "Regular Days" vs "Holidays".
- **Insight:** Median traffic on holidays is roughly 50% lower than regular days, indicating a massive reduction in commuter load.

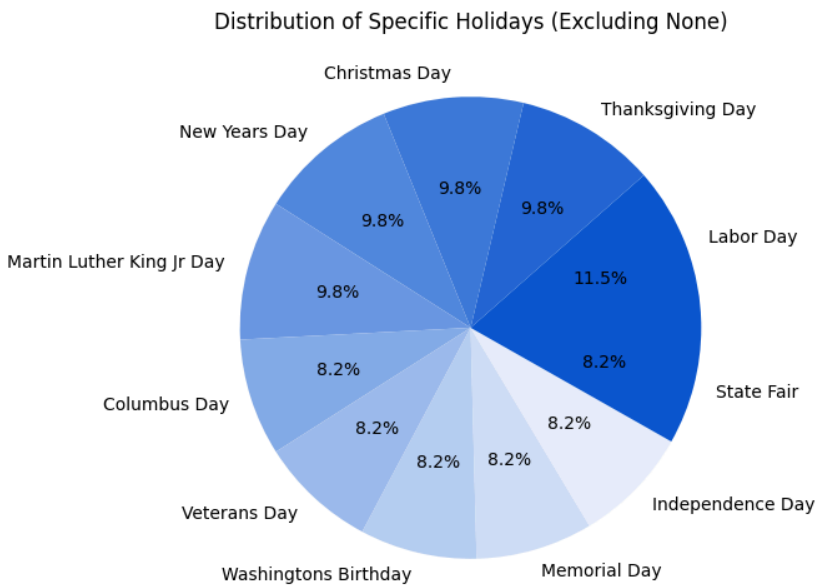
Graph 12: Clustered Heatmap (Hour vs Day)

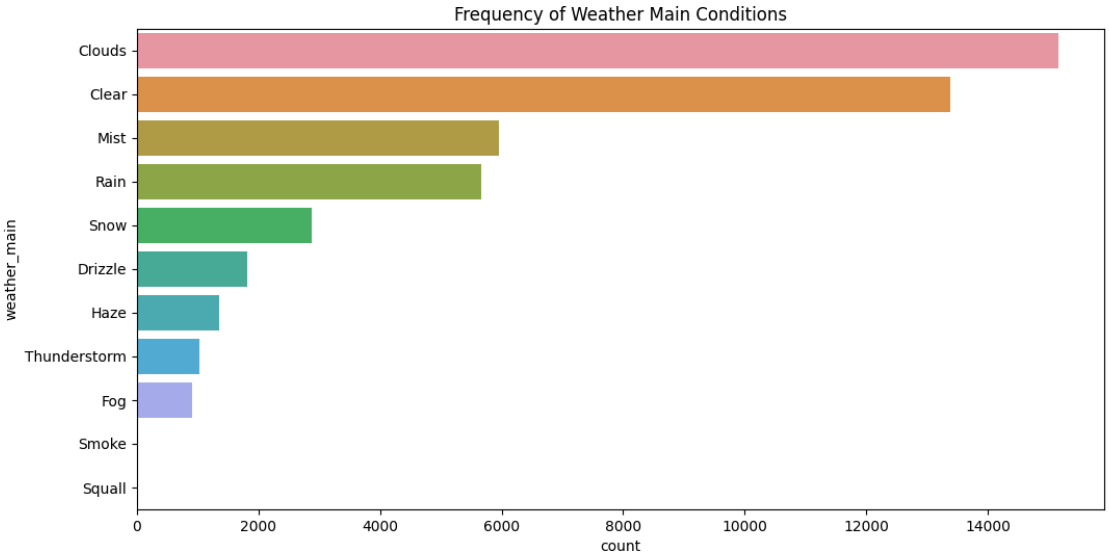
- **Description:** Intensity map showing the "hot zones" of traffic.
- **Insight:** High-intensity clusters are concentrated between 06:00-09:00 and 15:00-18:00 on weekdays only.

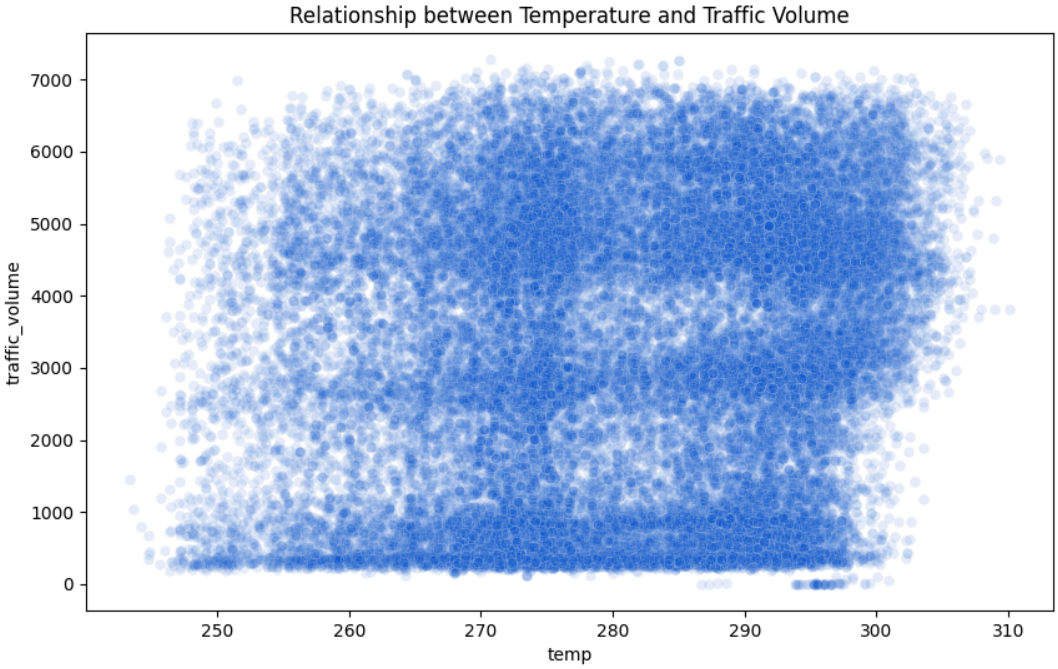
3. Final Output Section

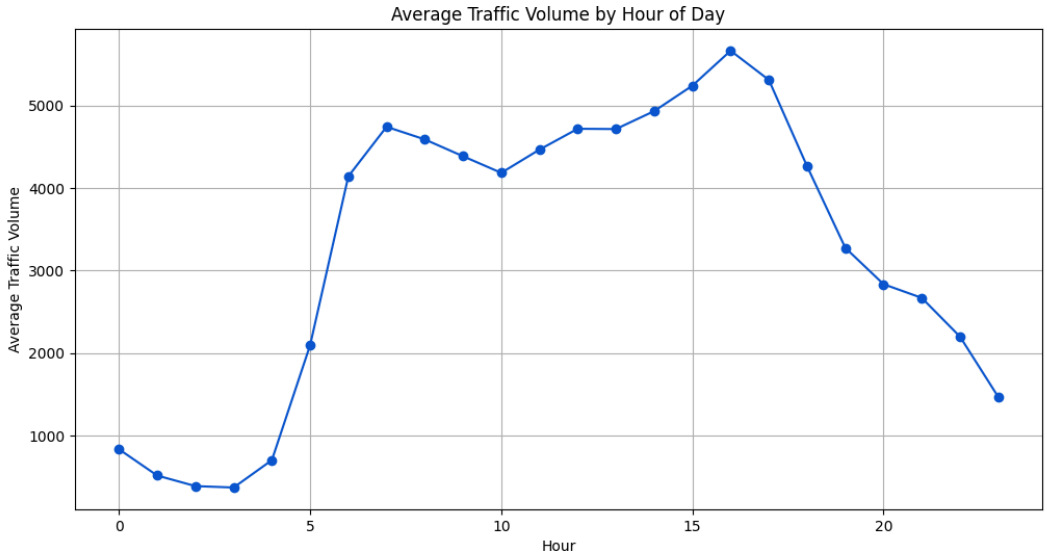
The analysis concludes that **Time (Hour/Day)** is a significantly better predictor of traffic volume than **Weather**. While extreme weather causes specific anomalies, the rhythmic nature of the work-week dominates interstate usage. The evening rush hour (16:00) represents the absolute

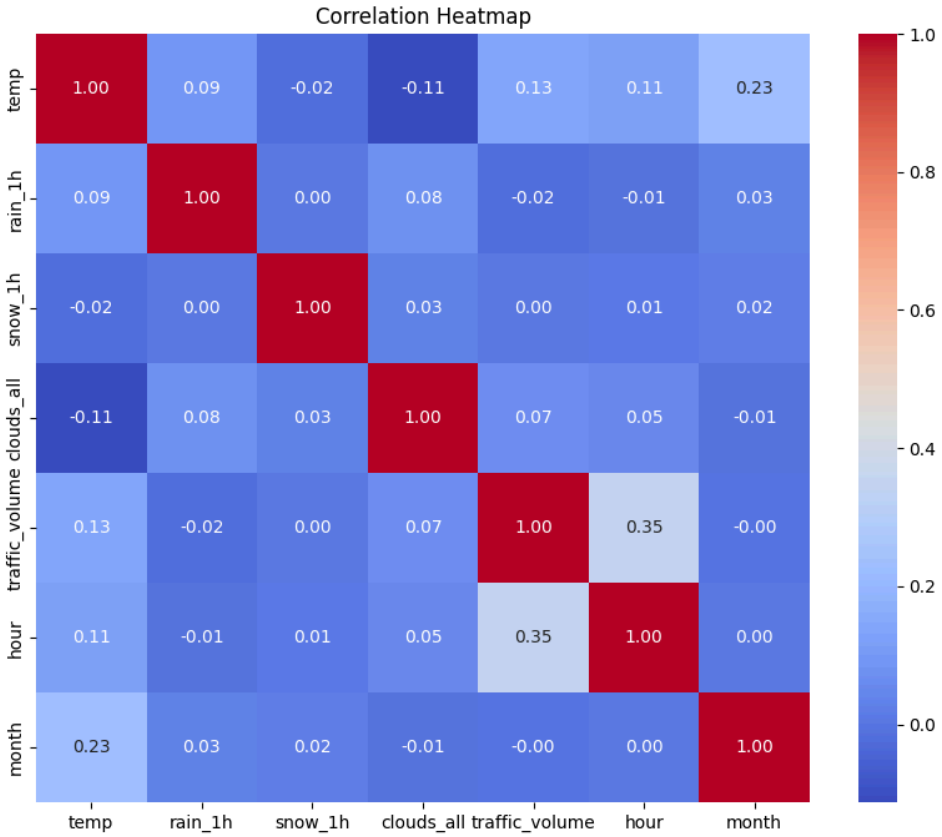
peak of infrastructure stress.

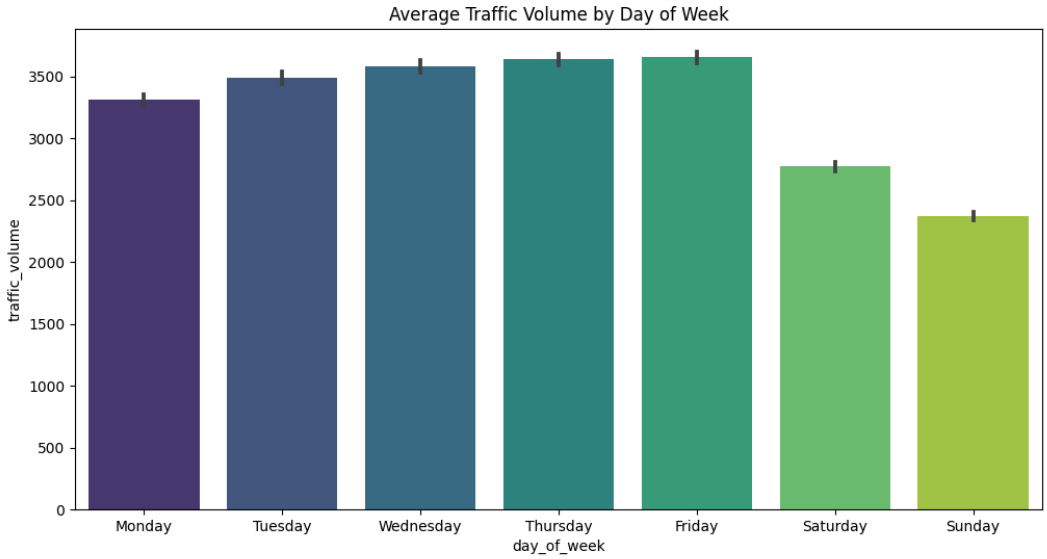


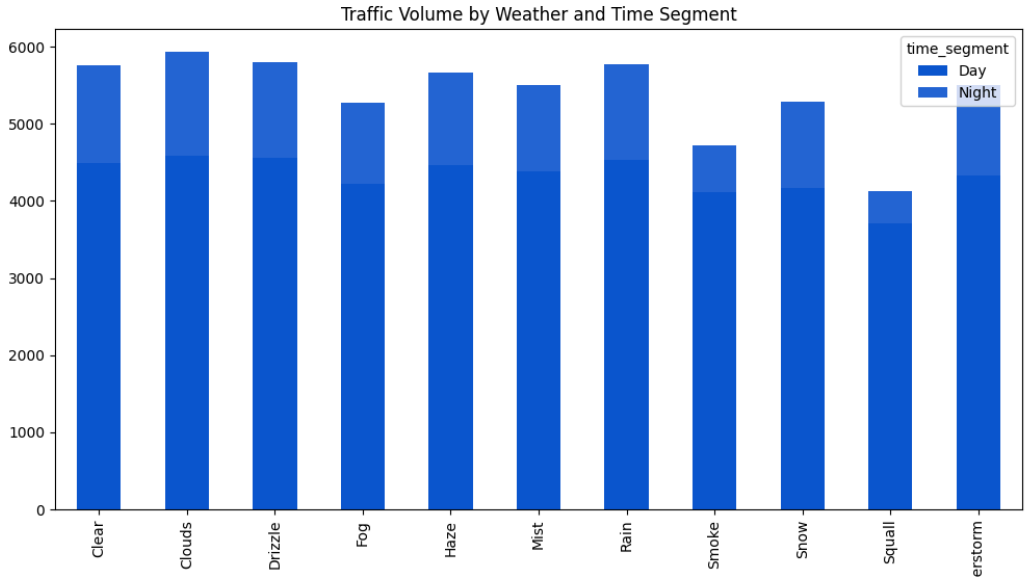

Frequency of Weather Main Conditions
Clouds
Clear
Mist
Rain
Snow
Drizzle
Haze
Thunderstorm
Fog
Smoke
Squall
weather_main
0
2000
4000
6000
8000
10000
12000
14000
count


Relationship between Temperature and Traffic Volume
7000
6000
5000
4000
3000
2000
1000
0
traffic_volume
250
260
270
280
290
300
310
temp


Average Traffic Volume by Hour of Day
Average Traffic Volume
Hour
1000
2000
3000
4000
5000
0
5
10
15
20


Correlation Heatmap
temp
rain_1h
snow_1h
clouds_all
traffic_volume
hour
month
1.00 0.09 -0.02 -0.11 0.13 0.11 0.23
0.09 1.00 0.00 0.08 -0.02 -0.01 0.03
-0.02 0.00 1.00 0.03 0.00 0.01 0.02
-0.11 0.08 0.03 1.00 0.07 0.05 -0.01
0.13 -0.02 0.00 0.07 1.00 0.35 -0.00
0.11 -0.01 0.01 0.05 0.35 1.00 0.00
0.23 0.03 0.02 -0.01 -0.00 0.00 1.00
1.0
0.8
0.6
0.4
0.2
0.0


Average Traffic Volume by Day of Week
traffic_volume
0
500
1000
1500
2000
2500
3000
3500
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
day_of_week


Traffic Volume by Weather and Time Segment
time_segment
Day
Night
0
1000
2000
3000
4000
5000
6000
Clear
Clouds
Drizzle
Fog
Haze
Mist
Rain
Smoke
Snow
Squall
erstorm